Target validation paths — pdac-recurrent-cldn18-2-kras-g12r-tp53-y220c-p25m

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Everything known about this tumor comes from a December 2025 resection specimen; the mass that appeared in July 2026 has never been sampled, and ten cycles of FOLFIRINOX sit between the two. So the workup here starts one step earlier than usual: before any target can be validated, the tumor itself has to be. One EUS-guided core pass supplies the histology, the CLDN18.2 re-stain, the DNA and RNA sequencing, and most tissue-gated trial screens, and the restaging read that travels with it decides access on its own: the daraxonrasib expanded-access protocol (NCT07573215) and the RASolute 303 trial (NCT07491445) both require confirmed metastatic disease, and RASolute 303 additionally requires consent within 6 weeks of that diagnosis. Two essential items need no tissue at all and should go out this week: the dedicated germline panel and a graded neuropathy examination. If the biopsy does not confirm recurrent pancreatic adenocarcinoma carrying these features, this report has no within-scope recommendations, and the next conversation about standard care is the treating team's, not Libby's.

The recurrence itself

The biopsy is the load-bearing order. Request at least three cores so one goes to a paraffin block for IHC and the rest to molecular; the likely vascular invasion is a route question rather than a barrier, since EUS from the stomach or duodenum usually avoids the vessels a percutaneous approach cannot. If the interventional team judges the lesion unsafe to sample, plasma ctDNA is the fallback and should be ordered in parallel rather than after. Three companions are essential and independent of tissue: a graded neurologic examination for peripheral sensory neuropathy (CTCAE v5 with the EORTC QLQ-CIPN20), because ten cycles of FOLFIRINOX is a large cumulative oxaliplatin exposure, no grade appears anywhere in the record, and most protocols exclude at grade 2 or higher; baseline CBC, comprehensive metabolic panel with bilirubin and albumin, creatinine with eGFR, coagulation studies, and a CA 19-9 that will only mean something later if drawn now; and pancreas-protocol cross-sectional imaging defining vascular involvement, biopsy route, local-only versus metastatic extent, and a RECIST-measurable target lesion. A fitness screen (fecal elastase-1, fat-soluble vitamins, HbA1c, biliary anatomy and stent status) rounds this out: enzyme replacement is cheap, and the weight loss it prevents is what later costs patients trial eligibility.

CLDN18.2

The 60% 2+/3+ result on file was produced by an assay the report does not name, on tissue from a lesion that is not the one being treated. Re-staining with the VENTANA CLDN18 (43-14A) RxDx assay puts the number on the scale every threshold conversation uses, including the sponsor assays for ASP2138 (NCT05365581) and the satri-cel program (CT041, NCT04404595). Just as essential: stain the recurrence biopsy and the December 2025 resection block side by side in one laboratory, because CLDN18.2 expression is discordant between primary and metastatic lesions in a substantial minority of gastric cases (PMID 40200874), loss under zolbetuximab treatment has been documented (PMID 41854734), and a recurrence-only stain that comes back lower cannot distinguish real antigen loss from inter-laboratory scoring drift. Ask the laboratory to report the intensity distribution at 1+, 2+, and 3+ separately rather than a single positive-or-negative call: the thresholds differ by program (>=75% at 2+/3+ for zolbetuximab in SPOTLIGHT and GLOW, >=40% at >=2+ for satri-cel, which the primary's 60% clears), and 60% sits exactly in the band where that distinction decides which door is open. Request the December 2025 block from the operating hospital early; retrieval is often the slow step.

KRAS G12R

Comprehensive tumor NGS on recurrent-lesion tissue, DNA plus RNA, is essential. The codon-level call is what separates a pan-RAS(ON) route such as daraxonrasib (RASolute 302, NCT06625320; RASolute 303, NCT07491445) from the G12C- and G12D-selective drugs that cannot work here, and re-sequencing tests whether the driver persisted through ten cycles of FOLFIRINOX. RNA rides along so fusion status ends up established rather than unaddressed. The plasma ctDNA panel reporting KRAS codon 12 and TP53 Y220C with variant allele fractions is the second essential item: it answers in about a week, it is the only route to a current genotype if the biopsy proves unsafe, and several mutant-KRAS vaccine protocols (ELI-002 7P, NCT05726864 and NCT07671339) use ctDNA positivity as an entry criterion, so it can open a door rather than merely inform. A negative plasma result in a low-shedding pancreatic lesion does not overturn tissue findings; that is the main way this assay gets misread. Co-alteration profiling (CDKN2A, SMAD4, ARID1A, MYC, KRAS allelic imbalance) refines what to expect from RAS-pathway inhibition and reads out on the same sequencing at no extra cost.

TP53 Y220C

PYNNACLE (NCT04585750) asks only that the tumor carry a TP53 Y220C mutation, but the companion rezatapopt protocols spell out what counts: CLIA-certified testing with a variant allele fraction above 2% (NCT06616636), tissue or liquid both acceptable (NCT07372625). The December 2025 tissue result probably satisfies this on its face; what the sponsor will want confirmed is that the variant is still present in current disease, so the cheapest version of this row is a phone call and a PDF before anything new is ordered. Two refinements are worth requesting rather than ordering: a purity-corrected variant allele fraction with TP53 copy number and LOH status on the recurrence NGS, which informs how much a Y220C reactivator can plausibly do, and, only once such a drug is actually started, serial plasma TP53 sequencing, because acquired secondary TP53 alterations arising in cis with Y220C drive on-target resistance in this class (PMID 41504628, with commentary at PMID 41918356) and show up in plasma before imaging turns.

BRCA and homologous recombination

The "no incidental germline findings" line on file came off a tumor NGS report, which is not built for germline sensitivity, misses large rearrangements, and does not state which genes were covered. The dedicated germline multigene panel (BRCA1, BRCA2, PALB2, ATM, CDKN2A, STK11, and the mismatch-repair genes) is a blood draw NCCN recommends for every pancreatic adenocarcinoma patient, and a pathogenic BRCA1/2 or PALB2 result puts an approved PARP-inhibitor maintenance strategy on the table (POLO, PMID 31157963 and PMID 35834777) while also arguing for keeping platinum in the sequence. It needs no tissue and should not wait on the biopsy. Its high-priority companion is somatic homologous-recombination gene coverage with a genomic-instability score on the recurrence NGS, which catches the somatic-only events a germline panel will miss and costs an order line rather than a new specimen.

MSS / TMB

Nothing here needs a standalone order. MSS and a TMB of 0.83 mut/Mb against a threshold of 10 are already at decision resolution and keep single-agent checkpoint blockade closed. The only caveat is that neither original assay platform was named, and panel-derived TMB is not interchangeable across platforms; that caveat bites near the cutoff, not an order of magnitude below it. If the recurrence is sequenced anyway, both values re-report free, and MMR IHC can ride on the same block.

KMT2C S2816fs

To be plain about it: no assay result would make this variant actionable today. No approved or late-phase therapy selects on KMT2C status in any tumor type. What a biallelic, clonal loss-of-function call would buy is entry to an epigenetic-agent basket trial that accepts COMPASS-complex alterations, if one opens to pancreatic patients. It rides on the recurrence NGS as a reporting detail and justifies no separate order.

Biomarkers measured, but not to decision resolution

Two results on file cannot carry the decisions that lean on them, and each is short in a different way.

- CLDN18.2. What exists is 60% of tumor cells at 2+/3+ on the December 2025 resection. Three things sit between that stain and an eligibility call: the number is below the >=75% line used in SPOTLIGHT and GLOW, the antibody clone and assay were never named so the result cannot be scored against the cutoff that assay defines, and the tissue predates ten cycles of chemotherapy and is not the lesion being treated. The 43-14A re-stain, ideally on recurrent tissue with the archival block run in the same batch, closes all three at once. The target call itself is not in doubt; the resolution of the measurement is.
- Homologous recombination / germline BRCA. What exists is "no incidental findings" from a tumor NGS report. That is not a germline test: it lacks germline sensitivity, misses large rearrangements, and does not say which genes it covered. The dedicated germline panel, plus somatic HR coverage with a genomic-instability score on the next tumor sequencing, closes the gap.

Where to order these assays

Assay	Provider	Decision gated	Contact
Comprehensive tumor NGS on recurrent-lesion tissue, DNA plus RNA, reporting codon-level KRAS genotype, TP53, homologous-recombination genes, fusions, MSI and TMB	**Caris Life Sciences (preferred) (MI Cancer Seek (whole exome plus whole transcriptome))**	**Pan-RAS(ON) trials for daraxonrasib (NCT06625320, NCT07491445) and rezatapopt via PYNNACLE (NCT04585750).**	**test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669**
Comprehensive tumor NGS on recurrent-lesion tissue	Foundation Medicine (FoundationOne CDx)	Pan-RAS(ON) trials for daraxonrasib (NCT06625320, NCT07491445) and rezatapopt via PYNNACLE (NCT04585750).	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Comprehensive tumor NGS on recurrent-lesion tissue	Tempus AI (Tempus xT (DNA) plus xR (RNA))	Pan-RAS(ON) trials for daraxonrasib (NCT06625320, NCT07491445) and rezatapopt via PYNNACLE (NCT04585750).	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
Comprehensive tumor NGS on recurrent-lesion tissue	NeoGenomics Laboratories (NeoTYPE Comprehensive Tumor Profile)	Pan-RAS(ON) trials for daraxonrasib (NCT06625320, NCT07491445) and rezatapopt via PYNNACLE (NCT04585750).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)
Plasma ctDNA comprehensive panel reporting KRAS codon 12 genotype and TP53 Y220C with variant allele fractions	**Guardant Health (preferred) (Guardant360 CDx)**	**Genotype for daraxonrasib and rezatapopt when tissue is not obtainable; ctDNA-positivity entry criterion for ELI-002 7P protocols.**	**test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887**

Plasma ctDNA comprehensive panel	Foundation Medicine (FoundationOne Liquid CDx)	Genotype for daraxonrasib and rezatapopt when tissue is not obtainable; ctDNA-positivity entry criterion for ELI-002 7P protocols.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Plasma ctDNA comprehensive panel	Tempus AI (Tempus xF+ (plasma))	Genotype for daraxonrasib and rezatapopt when tissue is not obtainable; ctDNA-positivity entry criterion for ELI-002 7P protocols.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
Plasma ctDNA comprehensive panel	Caris Life Sciences (Caris Assure (blood))	Genotype for daraxonrasib and rezatapopt when tissue is not obtainable; ctDNA-positivity entry criterion for ELI-002 7P protocols.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669
CLDN18.2 IHC by the VENTANA CLDN18 (43-14A) RxDx assay, reporting percent of viable tumor cells with moderate-to-strong (2+/3+) membranous staining	**NeoGenomics Laboratories (preferred) (Claudin 18 FDA (VYLOY), VENTANA CLDN18 (43-14A) RxDx Assay)**	**Off-label zolbetuximab and every CLDN18.2 trial that scores against the 43-14A assay (ASP2138 NCT05365581, CT041 NCT04404595).**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)**
CLDN18.2 IHC (43-14A), including the paired recurrence-plus-archival stain and the 1+/2+/3+ tier report	Labcorp Oncology (Claudin 18 IHC (test 452390))	Off-label zolbetuximab and every CLDN18.2 trial that scores against the 43-14A assay (ASP2138 NCT05365581, CT041 NCT04404595).	test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167
CLDN18.2 IHC (43-14A), including the paired recurrence-plus-archival stain and the 1+/2+/3+ tier report	Mayo Clinic Laboratories (CLD18: Claudin 18 (CLDN18) (43-14A), semi-quantitative IHC)	Off-label zolbetuximab and every CLDN18.2 trial that scores against the 43-14A assay (ASP2138 NCT05365581, CT041 NCT04404595).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Dedicated germline multigene panel on blood or saliva covering BRCA1, BRCA2, PALB2, ATM, CDKN2A, STK11 and the mismatch-repair genes	**Myriad Genetics (preferred) (MyRisk Hereditary Cancer with BRACAnalysis CDx)**	**PARP-inhibitor maintenance (olaparib) eligibility, platinum re-challenge weighting, and family testing.**	**test info · 322 North 2200 West, Salt Lake City, UT 84116 · 800-469-7423**
Dedicated germline multigene panel	Ambry Genetics (CustomNext-Cancer / PancNext)	PARP-inhibitor maintenance (olaparib) eligibility, platinum re-challenge weighting, and family testing.	test info · One Enterprise, Aliso Viejo, CA 92656 · 866-262-7943
Dedicated germline multigene panel	GeneDx	PARP-inhibitor maintenance (olaparib) eligibility, platinum re-challenge weighting, and family testing.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 888-729-1206
Dedicated germline multigene panel	Labcorp Genetics (formerly Invitae) (Multi-Cancer Panel)	PARP-inhibitor maintenance (olaparib) eligibility, platinum re-challenge weighting, and family testing.	test info · 1400 16th Street, San Francisco, CA 94103 · 800-436-3037
TP53 p.Y220C confirmation on a CLIA-certified NGS test (tumor tissue or plasma) reporting the variant at codon level with variant allele fraction	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Rezatapopt via PYNNACLE (NCT04585750).**	**test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639**

TP53 p.Y220C confirmation	Guardant Health (Guardant360 CDx)	Rezatapopt via PYNNACLE (NCT04585750).	test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887
TP53 p.Y220C confirmation	Tempus AI (Tempus xT (DNA) plus xR (RNA))	Rezatapopt via PYNNACLE (NCT04585750).	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
TP53 p.Y220C confirmation	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome))	Rezatapopt via PYNNACLE (NCT04585750).	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669
Somatic homologous-recombination gene coverage (BRCA1, BRCA2, PALB2, ATM, RAD51 paralogs) with a genomic-instability or LOH score on the recurrence NGS	**Caris Life Sciences (preferred) (MI Cancer Seek (whole exome plus whole transcriptome))**	**Somatic HRD evidence for off-label or trial PARP-inhibitor use and for keeping platinum in the sequence.**	**test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669**
Somatic HR gene coverage with genomic-instability score	Myriad Genetics (myChoice CDx (genomic instability score))	Somatic HRD evidence for off-label or trial PARP-inhibitor use and for keeping platinum in the sequence.	test info · 322 North 2200 West, Salt Lake City, UT 84116 · 800-469-7423
Somatic HR gene coverage with genomic-instability score	Foundation Medicine (FoundationOne CDx)	Somatic HRD evidence for off-label or trial PARP-inhibitor use and for keeping platinum in the sequence.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Somatic HR gene coverage with genomic-instability score	Tempus AI (Tempus xT (DNA) plus xR (RNA))	Somatic HRD evidence for off-label or trial PARP-inhibitor use and for keeping platinum in the sequence.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
Baseline CBC with differential, comprehensive metabolic panel with bilirubin and albumin, creatinine with calculated eGFR, coagulation studies, and CA 19-9	**Labcorp (preferred)**	**Organ-function eligibility thresholds shared by the trials under consideration, plus a baseline CA 19-9 for response assessment.**	**test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167**
Baseline organ-function labs and CA 19-9	Quest Diagnostics	Organ-function eligibility thresholds shared by the trials under consideration, plus a baseline CA 19-9 for response assessment.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 866-697-8378
Baseline organ-function labs and CA 19-9	Mayo Clinic Laboratories	Organ-function eligibility thresholds shared by the trials under consideration, plus a baseline CA 19-9 for response assessment.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Fecal elastase-1, fat-soluble vitamin levels (A, D, E), HbA1c, and review of biliary anatomy and stent status on the restaging scan	**Mayo Clinic Laboratories (preferred)**	**Nutritional and hepatobiliary fitness for systemic therapy and for trial performance-status and albumin thresholds.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710**

Exocrine and hepatobiliary assessment	Labcorp	Nutritional and hepatobiliary fitness for systemic therapy and for trial performance-status and albumin thresholds.	test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167
Exocrine and hepatobiliary assessment	Quest Diagnostics	Nutritional and hepatobiliary fitness for systemic therapy and for trial performance-status and albumin thresholds.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 866-697-8378

The EUS-guided biopsy, the restaging imaging, the neuropathy examination, and serial on-treatment ctDNA are performed through the treating center's interventional endoscopy, radiology, and oncology services rather than a reference laboratory, so they carry no provider rows here; the serial TP53 resistance-monitoring draws, once relevant, use the same platforms as the plasma ctDNA panel above.

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
EUS-guided core needle biopsy of the pancreatic body mass with histologic confirmation and tissue banked for downstream molecular and IHC work	Everything known about this tumor comes from a resection specimen taken before four more cycles of FOLFIRINOX, and the new mass has not been sampled, so its histology is presumed rather than established. One EUS-guided pass supplies the tissue that the CLDN18.2 re-stain, the DNA and RNA sequencing, and most tissue-gated trial screens all draw on. Skipping it means treating a radiographic diagnosis with a target call that may no longer hold.	interventional endoscopy at the treating center (no reference-lab order)	EUS-guided core biopsy; request at least three cores so one goes to a paraffin block for IHC and the rest to molecular

Comprehensive tumor NGS on recurrent-lesion tissue, DNA plus RNA, reporting codon-level KRAS genotype, TP53, genes, fusions, MSI and TMB	G12R is not touched by the G12C-selective drugs (sotorasib, adagrasib) or by G12D-selective agents, so the codon-level call is what separates a pan-RAS(ON) route such as daraxonrasib from drugs that cannot work here; RASolute 302 (NCT06625320) asks for documented RAS status at codon 12, 13, or 61. Re-sequencing the recurrence also tests whether the driver persisted through ten cycles of FOLFIRINOX and catches anything acquired under that pressure. A DNA-only panel leaves fusion status unestablished rather than negative, which is why RNA should ride along.	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669	FFPE block or 10-15 unstained slides from the recurrent-lesion biopsy; archival resection block acceptable if the biopsy fails
Plasma ctDNA comprehensive panel reporting KRAS codon 12 genotype and TP53 Y220C with variant allele fractions	If the lesion turns out to be unsafe to biopsy, plasma is the only route to a current genotype, and it answers within about a week rather than a month. Several mutant-KRAS vaccine protocols in this space use ctDNA positivity as an entry criterion (ELI-002 7P, NCT05726864 phase 1, NCT07671339), so the result can gate access rather than merely inform it. A negative plasma result in a low-shedding pancreatic lesion does not overturn the tissue findings, which is the main way this assay is misread.	Guardant Health (Guardant360 CDx) · test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887	10-20 mL whole blood; no tissue and no procedure

CLDN18.2 IHC by the VENTANA CLDN18 (43-14A) RxDx assay, reporting percent of viable tumor cells with moderate-to-strong (2+/3+) membranous staining	The existing 60% 2+/3+ result was produced by an assay the report does not name, and the >=75% cutoff that defines zolbetuximab eligibility in SPOTLIGHT and GLOW is specific to the 43-14A companion assay, so an unnamed clone cannot be scored against it. Re-staining with the FDA assay puts the number on the scale that every threshold conversation uses, including the sponsor assays for ASP2138 (NCT05365581) and CT041 (NCT04404595), which require central or protocol-specified IHC. Without it the whole CLDN18.2 strategy rests on a percentage that cannot be compared to any published threshold.	NeoGenomics Laboratories (Claudin 18 FDA (VYLOY), VENTANA CLDN18 (43-14A) RxDx Assay) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)	FFPE block, or one H&E plus 2-3 positively charged unstained slides at 4-5 microns with more than 100 viable tumor cells
Paired CLDN18.2 IHC on the recurrent-lesion biopsy and the December 2025 resection block, stained and scored side by side in one laboratory	CLDN18.2 expression is discordant between primary and metastatic lesions in a substantial minority of gastric cases (PMID 40200874), and loss of expression under zolbetuximab treatment has been documented in a reported case (PMID 41854734), so a stain from a pre-adjuvant resection is weak evidence about a lesion that emerged after ten cycles of chemotherapy. Running both specimens in the same laboratory in the same batch is what separates real biological change from inter-laboratory scoring drift. If only the recurrence is stained and comes back lower, there will be no way to tell which of the two explanations applies.	NeoGenomics Laboratories (Claudin 18 FDA (VYLOY), VENTANA CLDN18 (43-14A) RxDx Assay) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)	unstained slides from both blocks; request the December 2025 resection block from the operating hospital early, since retrieval is often the slow step

Dedicated germline multigene panel on blood or saliva covering BRCA1, BRCA2, PALB2, ATM, CDKN2A, STK11 and the mismatch-repair genes	The 'no incidental germline findings' line came off a tumor NGS report, which is not built for germline sensitivity, misses large rearrangements, and does not state which genes were covered. NCCN recommends germline testing for every patient with pancreatic adenocarcinoma, and a pathogenic BRCA1/2 or PALB2 variant puts an approved PARP-inhibitor maintenance strategy on the table in this tumor type (POLO, PMID 31157963 and PMID 35834777) as well as arguing for keeping platinum in the sequence. It is a blood draw with no procedure attached, so it should go out now rather than wait on tissue.	Myriad Genetics (MyRisk Hereditary Cancer with BRACAnalysis CDx) · test info · 322 North 2200 West, Salt Lake City, UT 84116 · 800-469-7423	5-10 mL whole blood in EDTA, or a saliva kit
TP53 p.Y220C confirmation on a CLIA-certified NGS test (tumor tissue or plasma) reporting the variant at codon level with variant allele fraction	PYNNACLE (NCT04585750) states only that the tumor must carry a TP53 Y220C mutation, but the companion rezatapopt protocols spell out what counts: CLIA-certified local testing with variant allele fraction above 2% (NCT06616636), and tissue-based or liquid-biopsy NGS both acceptable (NCT07372625). The December 2025 tissue result probably satisfies this on its face; what the sponsor will want confirmed is that the variant is still present in current disease. Turning up at screening with only a pre-adjuvant report risks a re-test delay at best.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	no new specimen if the existing CLIA report is accepted; otherwise rides on the recurrence NGS or the plasma ctDNA draw

Graded neurologic examination for peripheral sensory neuropathy (CTCAE v5) with a patient-reported instrument (EORTC QLQ-CIPN20) at baseline	Ten total cycles of FOLFIRINOX is a large cumulative oxaliplatin exposure and no neuropathy grade appears anywhere in the record, yet most trial protocols exclude grade 2 or higher peripheral neuropathy and every platinum or taxane re-exposure decision turns on it. A documented baseline grade also gives the toxicity side of the 0.7 efficacy weight something real to sit on. Without it, both the ranking and the screening conversation are guessing about the one toxicity this patient is most likely to already carry.	treating oncology clinic (examination plus questionnaire; no lab order)	none; clinic examination and a questionnaire
Baseline CBC with differential, comprehensive metabolic panel with bilirubin and albumin, creatinine with calculated eGFR, coagulation studies, and CA 19-9	No laboratory values were supplied at intake, so marrow, renal, and hepatic reserve after ten cycles of chemotherapy and a pancreatic resection are entirely unknown, and these thresholds appear in the eligibility criteria of essentially every trial under consideration. A baseline CA 19-9 is also needed now if it is going to mean anything as a response measure later. Screening failures on labs that were never checked are the most avoidable way to lose weeks here.	Labcorp · test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167	one routine blood draw; can share the germline panel and ctDNA venipuncture

Pancreas-protocol CT of chest, abdomen and pelvis (or MRI abdomen) defining vascular involvement, biopsy route, and local-only versus metastatic extent, with target lesions identified	The recurrence is described as a body mass with likely vascular invasion, which leaves open both whether disease is confined to the pancreatic bed and whether a measurable target lesion exists; most trials require the latter. The same scan tells the interventional team whether an EUS or percutaneous route to the mass is safe. Local-only disease also keeps radiation and locoregional options in the conversation that widely metastatic disease would close.	radiology at the treating center (no reference-lab order)	none; imaging only
Somatic gene coverage (BRCA1, BRCA2, PALB2, ATM, RAD51 paralogs) with a genomic-instability or LOH score on the recurrence NGS	A germline panel will miss somatic-only BRCA and PALB2 events, which occur in pancreatic adenocarcinoma and carry the same platinum-sensitivity signal even though the olaparib label is written around germline status. A genomic-instability score adds a phenotypic read when no single gene is hit. This rides on sequencing already recommended for the recurrence, so it costs an order line rather than a new specimen.	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669	no additional tissue beyond the recurrence NGS specimen

Quantified CLDN18.2 expression tier: percent of tumor cells at 1+, 2+ and 3+ membranous intensity, reported against each program's threshold	A single positive-or-negative call is useless here because the thresholds differ by program: >=75% at 2+/3+ for zolbetuximab (SPOTLIGHT, GLOW), and >=40% at >=2+ intensity for satri-cel, which the primary's 60% clears. Asking the laboratory to report the intensity distribution rather than one number lets each protocol be checked against its own criterion without re-staining. A 60% result sits in the band where this distinction decides which door is open.	NeoGenomics Laboratories (Claudin 18 FDA (VYLOY), VENTANA CLDN18 (43-14A) RxDx Assay) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)	no additional tissue; a reporting request on the CLDN18.2 re-stain
Fecal elastase-1, fat-soluble vitamin levels (A, D, E), HbA1c, and review of biliary anatomy and stent status on the restaging scan	Exocrine insufficiency after pancreatic resection is common, under-treated, and drives the weight loss and hypoalbuminemia that later cost patients trial eligibility and dose intensity; enzyme replacement is cheap and fixes it. New endocrine insufficiency after resection is frequent enough to check for. Biliary anatomy and any indwelling stent matter because cholangitis during systemic therapy is a common reason treatment stops.	Mayo Clinic Laboratories · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710	one stool sample plus routine blood; no procedure

TP53 Y220C variant allele fraction corrected for tumor purity, with TP53 copy number and LOH status, from the recurrence NGS	A Y220C reactivator only helps if the mutant protein is the dominant p53 species in most tumor cells, so a subclonal variant or a second inactivating hit on the other allele changes what the drug can plausibly do. In pancreatic adenocarcinoma TP53 alterations are typically clonal with LOH, so the expected answer supports the target rather than undercutting it. This does not gate PYNNACLE entry; it informs how much to expect.	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669	no additional tissue; a reporting request on the recurrence NGS
Serial plasma ctDNA TP53 sequencing during rezatapopt treatment to detect acquired secondary TP53 alterations in cis	Acquired secondary TP53 mutations arising in cis with Y220C drive clinical resistance to the Y220C reactivator (PMID 41504628, with accompanying commentary PMID 41918356), and they are detectable in plasma before imaging turns. Knowing that resistance is on-target rather than target-independent changes whether a next-generation p53 reactivator is worth pursuing. Only relevant once rezatapopt is actually started, which is why this sits below the eligibility rows.	Guardant Health (Guardant360 CDx) · test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887	10-20 mL whole blood per timepoint

Co-alteration profiling on the recurrence NGS: CDKN2A, SMAD4, ARID1A, MYC amplification, and KRAS allelic imbalance or amplification	KRAS allelic imbalance and amplification are among the mechanisms that blunt RAS-pathway inhibition, and SMAD4 loss carries a distinct metastatic phenotype in pancreatic adenocarcinoma that bears on how aggressively to sequence therapies. None of these findings would exclude a pan-RAS(ON) trial, so this refines expectation rather than eligibility. It reads out on sequencing already ordered.	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 888-979-8669	no additional tissue; included in the recurrence NGS report
MMR IHC (MLH1, PMS2, MSH2, MSH6) plus MSI and TMB re-read as part of the recurrence NGS; no standalone order	MSS status and a TMB of 0.83 mut/Mb against a threshold of 10 are already at decision resolution and close the tumor-agnostic pembrolizumab routes; nothing here needs a new order. The only caveat is that neither assay platform was named, and TMB from a small panel is not interchangeable across platforms, a caveat that bites near the cutoff rather than an order of magnitude below it. If the recurrence is biopsied and sequenced anyway, both values re-report at no extra cost and MMR IHC can ride on the same block.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 (Client Services, option 3)	no additional tissue; one unstained slide set if MMR IHC is added to an existing block

Characterization of KMT2C p.S2816fs as truncating and biallelic: variant allele fraction with tumor purity, KMT2C copy number and LOH, from the recurrence NGS	To be plain about it: no assay result would make this variant actionable today, because no approved or late-phase therapy selects patients on KMT2C status and the synthetic-lethal hypotheses remain preclinical. What a biallelic, clonal loss-of-function call would buy is entry to an epigenetic-agent basket trial that accepts COMPASS-complex alterations, if one is open and taking pancreatic patients. Nothing about this justifies a separate order or a separate specimen.	no separate order; a reporting request on the recurrence NGS	no additional tissue; a reporting detail on the recurrence NGS

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.